RESEARCH DESIGN CONVERSATION

# Decision Science × Socratic AI

Publication route • Experimental groups • Intervention timing
Behaviour vs computation • Bayesian models • Ethics & next steps

Working discussion with Martín | 13 October 2026

Proposals for discussion, not agreed protocol or verified AI effects

# Your 8-layer Socratic teammate architecture

The experiment is one operational layer inside the larger system. The design options below are alternative ways to test the same architecture.

**1 Experiment**
MAB/TBS events, trial timing, logs, outcomes

**2 Environment**
MAB hidden-state belief; TBS explicit constraints

**3 Participant**
Calibrated, uncertain, task-specific model

**4 Knowledge base**
Fixed ontology + versioned dynamic records

**5 Epistemic annotations**
Known / inferred / contested / unknown

**6 Reasoning**
Bayesian checks + alternative explanations

**7 LLM**
Grounded wording only, schema-constrained output

**8 Intervention policy**
Ask / explain / challenge / justified silence

## How the experiment fits

***Calibration builds Layer 3*** **(participant model).**
***Conditions manipulate Layers 6–8*** **by varying what the system knows about the task and person, and how the Socratic teammate intervenes.**

# Option 1: compact four-condition study

This is the simpler first experiment. It uses a common unassisted calibration and then compares four conditions.

**Common MAB + TBS calibration → behavioural & computational profile**

A

**No AI**
No assistance
Matched neutral waiting window

B

**General AI**
Generic guidance
Not task- or person-specific

C

**Task-aware AI**
Uses current task state
No calibrated person model

F

**Socratic teammate (task + person)**
Task state + calibrated person model
Question / challenge / justified silence

**What this design tests**
B vs A: any value of generic AI support
C vs B: added value of task awareness
F vs C: combined added value of personalisation + Socratic teammate mechanism

**What it does NOT isolate cleanly**
*F vs C does not separate Socratic delivery from personalisation*
Useful if the first question is practical effectiveness, not mechanism

**How it fits the 8-layer model**
A–C vary Layers 6–8 with no calibrated participant model
F adds the calibrated person model from Layer 3 into the intervention policy

# Option 2: expanded six-condition study

**Same common calibration → same participant-model starting point**

A

**No AI**
No assistance

B

**General AI**
Generic advice only

C

**Task-aware AI**
Task state only

D NEW

**Task-aware Socratic AI teammate**
Task state + Socratic delivery

E NEW

**Task-aware + personalised AI**
Task + calibrated person model

F

**Task-aware + personalised Socratic AI teammate**
Task + person + Socratic delivery

**Why add D and E?**
D isolates the Socratic teammate mechanism without personalisation
E isolates personalisation without Socratic delivery

**What improves?**
Cleaner mechanism tests
Clearer mapping from Layers 3, 6, 7 and 8 to outcomes

**Trade-off**
Larger sample, more implementation, more ethics and piloting work

# Four matched AI conditions answer the mechanism question

Keep the baseline conditions outside the factorial and pre-register the primary endpoints.

| | Conventional delivery | Socratic mechanism |
|---|---|---|
| Environment only (D) | 2 Context only | 3 Context only |
| Environment + person (E) | 4 Context + person | 5 Context + person |

## Socratic main effect

Average of (3–2) and (5–4).

## Personalisation main effect

Average of (4–2) and (5–3).

# Timing follows the information structure of each task

Preserve a five-second commitment period; standardise the extra exposure in every condition.

## MAB | new evidence is feedback

| Choice (Commitment) | Feedback | AI / neutral | Options Display |
|---|---|---|---|
| 5 s | original | 3–5 s | 5 s |

## TBS | new evidence is the displayed option pair

| Options Display + AI / neutral | Choice (Commitment) | Feedback |
|---|---|---|
| 3–5 s | 5 s | original |

**A. receives a neutral window; B/C/D receive the same fixed window. TBS options are visible in ALL conditions during it.**

*To specify whether choice is disabled in the preparation window, and log AI generation latency, reading exposure and early response attempts.*

# Behaviour is observed; computation is inferred

We need to keep these epistemic levels explicit in the intervention engine.

### 1 OBSERVATION

“Participant stays with Blue 1.”

The choice and reward are directly recorded.

### 2 ESTIMATION

“Model Q for Blue 1 is 0.2; another option is 0.7.”

Depends on fitted parameters and history.

### 3 HYPOTHESIS

“This persistence may not match the estimated values.”

Competing explanations remain plausible.

**Same immediate action ≠ same context, learned values, personal policy, intent or correct intervention.**

*No computational variable by itself reads a participant’s mind or proves a cognitive deficit.*

# Same visible choice, different model interpretation

MAB • SAME BANDIT CHOICE

## Both stay with Blue 1 after 0 reward

| | Person A | Person B |
|---|---|---|
| Q Blue 1 | 0.8 | 0.2 |
| Best alternative Q | 0.4 | 0.7 |
| Possible reading | Model-consistent | Model-inconsistent |

TBS • SAME RISKY CHOICE

## Same task state, different calibration

**Person A:** risky choice varies with target pressure.
**Person B:** risky choice remains frequent across low/high pressure.

A Bayesian policy model may assign different pressure-sensitivity estimates, with uncertainty.

*MAB Q values are fitted quantities; TBS individual risk-pressure policy is a proposed Bayesian extension, not an already fitted Chapter 5 model.*

# MAB: four key measures

From my Chapter 5 RL and ideal-observer work.

| Measure / calculation | What it can show | What it cannot show |
|---|---|---|
| 1. Learned option Q<br>$Q' = Q + \alpha(r - Q)$ | Model-estimated attractiveness/value of each arm before the next choice | Participant's conscious belief or true hidden reward probability |
| 2. Prediction error<br>$\delta = r - Q$ chosen | Whether feedback differs from model expectation; sign and size | Proof the participant noticed a change in contingencies |
| 3. Choice probability<br>softmax ($\beta Q$) | How compatible the observed choice is with fitted values | Subjective confidence, deliberation or true optimality |
| 4. Bayesian state belief<br>P (state \| outcome history) | Uncertainty that the colour-group contingency is in original vs swapped state | Actual hidden shift time, certainty of a state change |

*Observed behavioural counterparts: choice, reward, switch/stay, feedback-conditioned adjustments; dual learning rates allow $\alpha+$ and $\alpha-$.*

# MAB: a model state does not imply one action

One possible chain of reasoning, with alternative explanations preserved.

### Observed

Participant stays after an unrewarded outcome; several recent rewards were low.

### Computed

Q may be declining; estimated probability of contingency change rises. Both estimates remain uncertain.

### Socratic policy

Ask “What would make you reconsider?” only if grounded and useful; otherwise stay silent.

**MAB intervention: after feedback → fixed 3–5 s → next choice (5 s).**

*The AI must not receive the experiment runner’s actual reward schedule or actual volatility switch events.*

# TBS: four key measures

Task-state arithmetic is established; Bayesian participant-policy modelling is the proposed extension.

| Measure / calculation | What it can show | What it cannot show |
|---|---|---|
| 1. Target gap<br>G = target – current score | Remaining nominal score required at current trial | Whether target is reachable through current choices |
| 2. Target pressure<br>G / remaining opportunities | Demand per remaining choice as the horizon shrinks | That taking more risk is necessarily optimal |
| 3. Option expected value<br>EV = P(win) × gain | Average immediate points from displayed alternatives | Probability of final target attainment |
| 4. Bayesian policy sensitivity*<br>P(risky \| gap, horizon, history) | Uncertain estimate of how the person adjusts risky choices to context | Stable trait, intent, or evidence that AI help will work |

**Future model, not computed in my original TBS Step 02. Existing recorded behaviour includes safe/risky choices and shifts following gains/no gains.*

# TBS: current alternatives + prior policy can both matter

Illustrative task-state arithmetic, not participant data.

## What the task tells us

Target 930 | Score 450 | 2 trials left
Gap = 480 | Pressure = 240/trial

Safe: 80% × 100 → EV 80
Risky: 35% × 265 → EV 92.75

EV alone cannot choose the best target-attainment policy.

## What the person model adds

Both choose risky now.

A: calibration suggests strong responsiveness to target pressure.
B: calibration suggests weak responsiveness.

The Bayesian model must represent uncertainty and compare competing explanations.

**TBS timing: options visible for 3–5 s (all groups) → five-second commitment window.**

*Multiplier changes final-trial feasibility: required raw gain after a winning final trial = target / multiplier − prior score.*

# Three Bayesian roles, one Socratic teammate

Irena's task-specific inference supplies evidence; Martín's network assesses interpretations. A separate policy selects intervention or silence.

### MAB | EXISTING Bayesian ideal observer

Why before? Reward contingencies are hidden and may change. Bayesian inference was needed to estimate the environment.

Input → observed rewards + permitted task rules
Output → P(contingency state | reward history)

Also: fitted RL Q-values, prediction errors, switching

### TBS | PROPOSED Bayesian participant policy

Why not before? Probabilities, gains and constraints were shown. Original analysis calculated task state and behavioural adaptation directly.

Why now? Estimate uncertainty in how this person responds to pressure, horizon and outcomes.

Output → P(context-sensitive policy | choice history)

### Epistemic Bayesian reasoning network | pgmpy

Combines task evidence + uncertain person hypotheses → tests alternatives, evidential strength and likely usefulness

**Socratic intervention policy → ask / challenge / explain / remain silent**

Critical: posterior ≠ knowledge of thoughts; a high posterior does not automatically justify interruption. TBS Bayesian policy remains to be built and validated.